Experiment 4: Entropy, Cross-Entropy and KL Divergence

Aim

To calculate entropy, cross-entropy and KL divergence for observed weather probabilities and two forecasting models, verify the relationship between these measures, and demonstrate the asymmetry of KL divergence.

Mathematical Formulae

1. Entropy

$$H(P) = -\sum_i P_i \log_2 P_i$$

2. Cross-entropy

$$H(P, Q) = -\sum_i P_i \log_2 Q_i$$

3. KL divergence

$$D_{KL}(P\|Q) = \sum_i P_i \log_2 \frac{P_i}{Q_i}$$

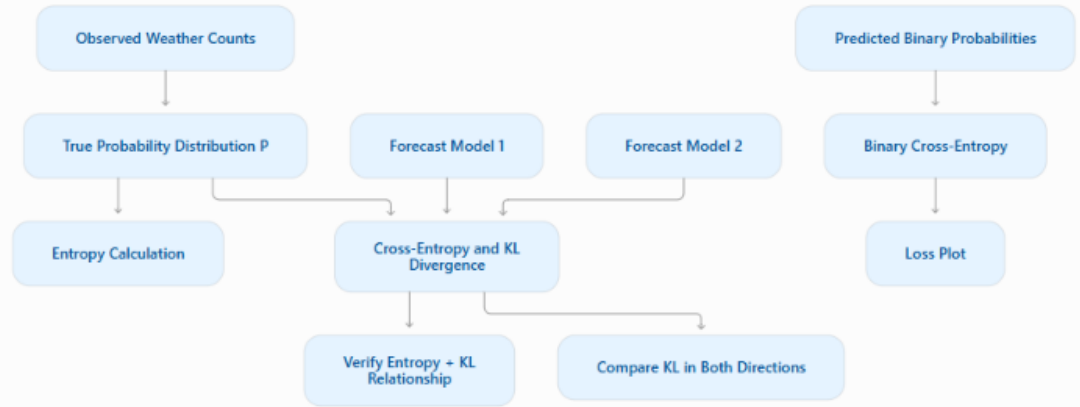
4. Relationship

$$H(P, Q) = H(P) + D_{KL}(P\|Q)$$

5. Binary cross-entropy

$$L = -[y \log_2 p + (1 - y) \log_2 (1 - p)]$$

System Architecture



Algorithm / Procedure

1. Define the observed weather counts for Sunny, Cloudy and Rainy days.
2. Convert the counts into probabilities.
3. Define probability distributions for two forecasting models.
4. Calculate the entropy of the observed distribution.
5. Compute cross-entropy and KL divergence for both models.
6. Verify the relationship between entropy, cross-entropy and KL divergence.
7. Calculate KL divergence in both directions to demonstrate asymmetry.
8. Plot the observed and predicted distributions.
9. Plot binary cross-entropy loss against predicted probability.

Python Program

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import entropy

# 1. Observed weather distribution over 30 days
true_counts = np.array([15, 10, 5])
P = true_counts / true_counts.sum()

# Two competing forecasting models
Q1 = np.array([0.50, 0.30, 0.20])
Q2 = np.array([0.70, 0.20, 0.10])

weather = ["Sunny", "Cloudy", "Rainy"]

print("True Distribution:", P)
print("Model 1:", Q1)
print("Model 2:", Q2)

# 2. Entropy of the true distribution
```

```
H = entropy(P, base=2)
print("\nEntropy of True Distribution:", round(H, 4), "bits")

# 3. Cross-entropy and KL divergence
def compare_model(P, Q, name):
    cross_entropy = -np.sum(P * np.log2(Q))
    kl = entropy(P, Q, base=2)

    print(f"\n{name}")
    print("Cross-Entropy:", round(cross_entropy, 4), "bits")
    print("KL Divergence:", round(kl, 4), "bits")
    print("Entropy + KL:", round(H + kl, 4), "bits")

    return cross_entropy, kl

ce1, kl1 = compare_model(P, Q1, "Model 1")
ce2, kl2 = compare_model(P, Q2, "Model 2")

# 4. Verify cross-entropy = entropy + KL divergence
print("\nVerification")
print("Model 1:", np.isclose(ce1, H + kl1))
print("Model 2:", np.isclose(ce2, H + kl2))

# 5. Demonstrate asymmetry of KL divergence
kl_pq = entropy(P, Q1, base=2)
kl_qp = entropy(Q1, P, base=2)

print("\nKL(P || Q1):", round(kl_pq, 4))
print("KL(Q1 || P):", round(kl_qp, 4))
print("Are they equal?", np.isclose(kl_pq, kl_qp))

# 6. Plot weather distributions
x = np.arange(3)
width = 0.25
```

```
plt.figure(figsize=(8, 5))
plt.bar(x - width, P, width, label="Observed")
plt.bar(x, Q1, width, label="Model 1")
plt.bar(x + width, Q2, width, label="Model 2")

plt.xticks(x, weather)
plt.ylabel("Probability")
plt.title("Observed vs Predicted Weather Distributions")
plt.legend()
plt.tight_layout()
plt.show()

# 7. Binary cross-entropy loss against predicted probability
eps = 1e-7
predicted = np.linspace(0.001, 0.999, 300)
predicted = np.clip(predicted, eps, 1 - eps)

# Loss when actual label is 1 and when it is 0
loss_y1 = -np.log2(predicted)
loss_y0 = -np.log2(1 - predicted)

plt.figure(figsize=(8, 5))
plt.plot(predicted, loss_y1, label="Actual label = 1")
plt.plot(predicted, loss_y0, label="Actual label = 0")
plt.xlabel("Predicted Probability of Class 1")
plt.ylabel("Binary Cross-Entropy Loss (bits)")
plt.title("Binary Cross-Entropy Loss")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```

Output

```
True Distribution: [0.5        0.33333333 0.16666667]
Model 1: [0.5 0.3 0.2]
Model 2: [0.7 0.2 0.1]

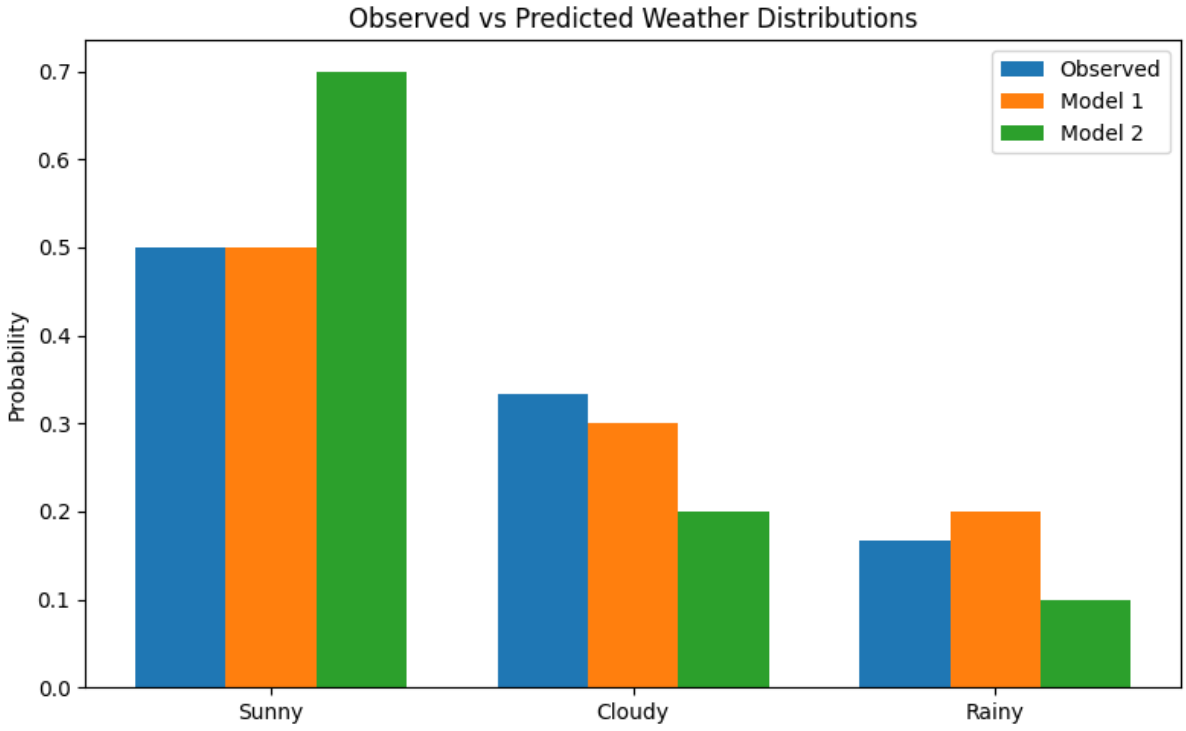
Entropy of True Distribution: 1.4591 bits

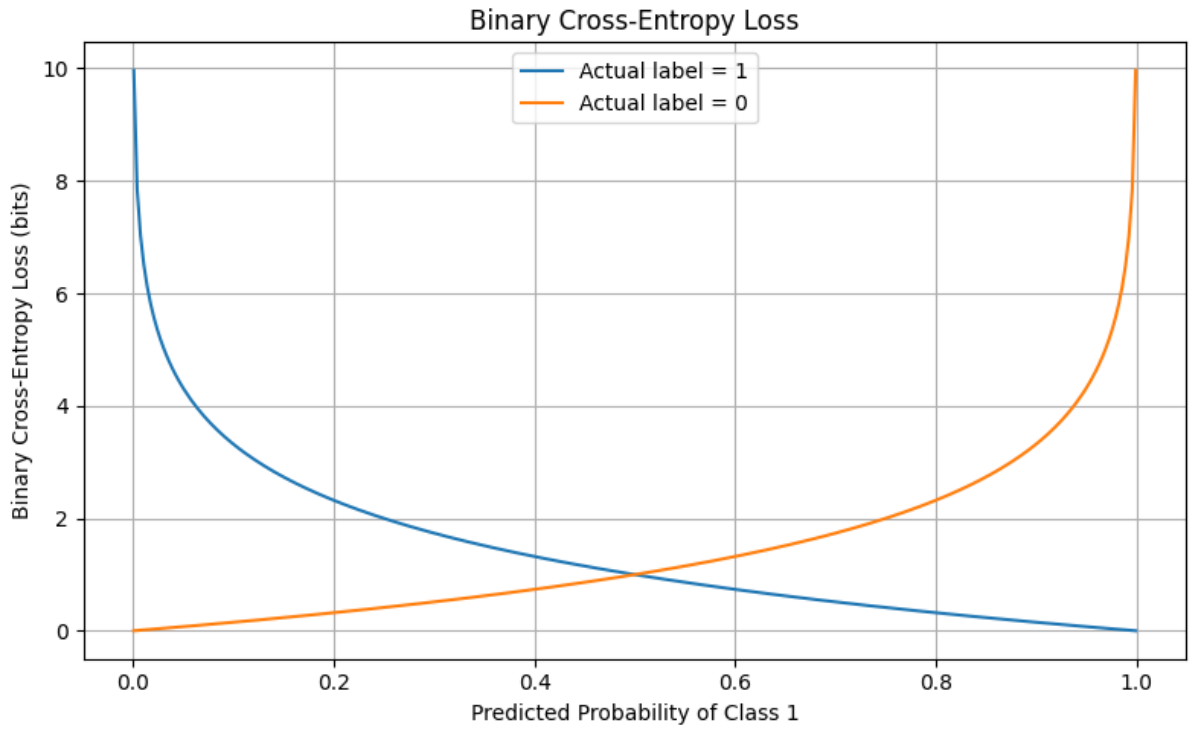
Model 1
Cross-Entropy: 1.466 bits
KL Divergence: 0.0068 bits
Entropy + KL: 1.466 bits

Model 2
Cross-Entropy: 1.5849 bits
KL Divergence: 0.1258 bits
Entropy + KL: 1.5849 bits

Verification
Model 1: True
Model 2: True

KL(P || Q1): 0.0068
KL(Q1 || P): 0.007
Are they equal? False
```





Result

Thus, entropy, cross-entropy and KL divergence were calculated for two forecasting models. The relationship between cross-entropy, entropy and KL divergence was verified, and the asymmetric nature of KL divergence was demonstrated.